A Certified Mode-Space Galerkin Method for Stochastic Partial Differential Equations, Uniform in Dimension

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Abstract

Validating a stochastic-PDE surrogate against a numerical reference is circular: two reasonable discretisations of the same white-noise-driven SPDE can disagree substantially on the identical noise realisation. We give a method whose error control never routes through a second solver: a mode-space Galerkin scheme for the linear bandlimited-Laplacian SPDE on $\mathbb{T}^d \times [0,T]$ with the computable a posteriori certificate $\|u_\theta - u_{\mathrm{ref}}\|_Y^2 \le C_1 \mathcal{L}_{\mathrm{RV}}(\theta) + C_2/L$ in $Y = L^2(0,T;H^1)$. In the Whittle–Matérn representation the problem decouples into independent per-mode problems: dimension enters only through the mode count, the coefficients are a closed-form $L \times L$ solve per mode, and both constants are uniform in d by construction. The substantive choice is the time basis, and it can fail silently: the natural Dirichlet sines stall at a non-decaying plateau (≈ 0.33 relative error at every L up to 1024) — detected by the certificate, invisible to compute — while the half-period basis (Dirichlet at $t=0$, Neumann at $t=T$) removes the obstruction and attains the optimal Karhunen–Loève rate $\approx 1.6/\sqrt{L}$ for the rough Ornstein–Uhlenbeck reference. Experiments confirm the designed dimension-independence (relative error unchanged within 8% as the mode set grows 444× from $d=2$ to $d=7$) and its transfer to real data (ERA5 temperature, SPX implied volatility); a stochastic Allen–Cahn extension drives the certified residual to 10^{-8} and attains the same $L^{-1/2}$ rate uniformly up to $d=5$, locating the nonlinear open problem in the approximation constant, not the certificate or the rate.

1 Introduction

How does one trust a surrogate solver for a stochastic PDE? The standard answer — validate against a numerical reference — carries a failure mode that is easy to state and, as it turns out, easy to measure. In the companion paper [2] we ran two independent, individually reasonable discretisations (ADI with upwind advection; MUSCL–minmod TVD finite volume) of the same white-noise-driven nonlinear SPDE on the *identical* noise realisation: they differ by 0.69 relative L^2 pathwise and by a factor 2.1 in field variance at the working grid, and for space–time white noise in two dimensions there is no continuum limit to arbitrate between them — the target is lattice-defined, and the lattice is solver-flavoured. Every reference-validated claim inherits that flavour silently. The structural remedy is an *a posteriori certificate*: a computable bound on the distance to the solution of the *declared* operator equation, with explicit constants, that never routes trust through a second solver.

This paper. We construct such a certificate, together with the method it certifies, for the linear bandlimited-Laplacian SPDE

$$\partial_t u = \Delta u + \sigma \dot{W}_{K_{\max}}, \qquad u(\cdot,0) = u_0, \tag{1}$$

on the torus $\mathbb{T}^d \times [0,T]$, where $\dot{W}_{K_{\max}}$ is a spatially bandlimited Gaussian white noise. In the Whittle–Matérn spectral representation $u(x,t) = \sum_{k\in\mathcal{H}_{K_{\max}}}[a_k(t)\cos(k\cdot x) + b_k(t)\sin(k\cdot x)]$, each coefficient a_k, b_k is an Ornstein–Uhlenbeck process, and the SPDE decouples into $J = |\mathcal{H}_{K_{\max}}|$ independent scalar problems in time. Working in this basis is the design decision from which everything follows: the spatial Laplacian becomes the diagonal multiplication by $-|k|^2$, the Galerkin system is an explicit $L \times L$ solve per mode, and — because the certificate is assembled per mode — *the spatial dimension enters only through the mode count.* Dimension-independence of the certified bound is not an empirical surprise to be discovered but a property of the architecture to be verified; our experiments verify it across a $444\times$ growth of the mode set. The price of the construction is the choice of a time basis $\{\psi_l(t)\}$ in which the per-mode equation is tested — and, as we show, that choice can fail silently.

Context. The method sits at the meeting point of two literatures. Residual-variational physics-informed networks (RVPINNs) minimise the residual in a discrete dual norm against a finite test space and yield reliable, efficient a posteriori estimators [6,7], of the form $\|u_\theta - u_{\mathrm{ref}}\|_V \le C\,\|R(u_\theta)\|_{V_h^*}$ with C an inf-sup constant; that literature concentrates on deterministic PDEs in $d \le 3$. Spectral (mode-space) representations for SPDE surrogates were developed in the distributional-regulariser line [2–4], whose theoretical guarantee is a Mishra–Molinaro-style [5] law-distance (W_2) bound with no path-level control; spectral-domain networks have also been proposed for deterministic high-dimensional PDEs [8], without per-instance error control. A priori error bounds for physics-informed and operator learning are given by De Ryck and Mishra [1]; the present certificate is the complementary a posteriori object — evaluated on the computed field rather than guaranteeing that a good one exists — and its constant is the computable, sharp realisation of their abstract stability constant. For the linear SPDE treated here no network and no training are involved at all: the certified object is a closed-form Galerkin solve, and the neural parameterisations of the companion papers appear only in the nonlinear extension of Section 5.

1.1 The time basis can fail silently: plateau and cure

A natural choice for the time basis is the standard Dirichlet sine basis $\psi_l(t) = \sqrt{2/T}\sin(\pi l t/T)$, $l = 1,\ldots,L$, which vanishes at both endpoints. With this basis the trial-space ansatz $a_{\theta,k}(t) = a_{0,k}e^{-|k|^2 t} + \sum_l \alpha_{k,l}\psi_l(t)$ satisfies $a_{\theta,k}(T) = a_{0,k}e^{-|k|^2 T}$ exactly, by construction; it cannot represent the SPDE's terminal noise accumulation $x_k(T) = \sigma\int_0^T e^{-|k|^2(T-s)}\,dB(s)$. The Galerkin formulation in this basis produces a linear system that admits a unique solution, but the resulting coefficients α *differ from the L^2-projection coefficients* of the true OU process by a tail-correction term that does not vanish as L grows. Empirically, the relative $L^2(0,T;H^1)$ error stalls at a plateau of ≈ 0.33 regardless of L, up to $L = 1024$ (Section 4.2) — a manifestation of the non-self-adjointness of the parabolic operator $\partial_t + |k|^2\,\mathrm{I}$ under this restricted test space. No amount of compute repairs it, and without an error instrument the failure is invisible: the Galerkin system remains well-posed and its residual small in the wrong norm. The certificate is precisely the instrument that exposes it.

The cure is architectural: replace the Dirichlet sine basis with the *half-period sine basis*

$$\phi_l(t) = \sqrt{2/T}\,\sin\!\left(\frac{(2l-1)\pi t}{2T}\right), \qquad l = 1,2,\ldots,L, \tag{2}$$

which satisfies $\phi_l(0) = 0$ but $\phi_l(T) = (-1)^{l-1}$ — free at the terminal time. These are the eigenfunctions of $-\partial_t^2$ with Dirichlet at $t=0$ and Neumann at $t=T$, the natural boundary conditions for the parabolic problem. With matched boundary conditions the Galerkin error attains the *optimal*

Karhunen–Loève rate for the rough (Hölder-$\frac{1}{2}$) OU reference, rel_err $\approx 1.6/\sqrt{L}$, uniformly across $d \in \{2, 3, 5, 7\}$ (Section 4.3). The corresponding certified bound is

$$\|u_\theta - u_{\text{ref}}\|_Y^2 \;\le\; C_1 \, \mathcal{L}_{\text{RV}}(\theta) + \frac{C_2}{L}, \tag{3}$$

with C_1 from the discrete inf-sup constant of the matched Galerkin formulation and C_2 from the best-approximation error of the OU process in the half-period basis. Both constants are uniform in d.

1.2 Contributions

1. **A certified mode-space Galerkin method, uniform in d by construction** (Sections 2, 3). Trial/test bases that make the Gram matrix block-diagonal in k and the per-mode time operator an explicit $L \times L$ system; the certificate $\|u_\theta - u_{\text{ref}}\|_Y^2 \le C_1 \mathcal{L}_{\text{RV}} + C_2/L$ assembles per mode, so d enters only through the mode count. Closed-form solve; no autograd, no FFT, no training.
2. **A silent structural failure, diagnosed and cured** (Section 4.2). The natural Dirichlet time basis stalls at a non-decaying plateau (≈ 0.33 up to $L = 1024$); the root cause is identified at the coefficient level (Galerkin $\neq L^2$ projection, non-vanishing tail correction); the half-period basis removes the obstruction and attains the optimal Karhunen–Loève rate $\approx 1.6/\sqrt{L}$ for the rough reference.
3. **Verification of the designed dimension-independence, synthetic and real** (Sections 4.3, 4.4). Across $d \in \{2, 3, 5, 7\}$ (J growing $444\times$) the relative error is unchanged within 8% at every L; on ERA5 temperature and SPX implied-volatility data the same uniformity holds, with the absolute rate set by the data's roughness and noise floor.

1.3 Roadmap

Section 2 describes the mode-space architecture, the half-period basis, and the closed-form linear solve. Section 3 states and proves Theorem 2. Section 4 reports the empirical validation (4.2 the negative result; 4.3 the uniform-in-d verification). Section 5 discusses extensions to nonlinear SPDEs.

2 Method

2.1 The bandlimited-Laplacian SPDE and its mode-space solution

We consider the linear bandlimited-Laplacian SPDE (1) on $\mathbb{T}^d \times [0, T]$, where the noise is bandlimited: $\dot{W}_{K_{\max}}(x, t) = \sum_{k \in \mathcal{H}_{K_{\max}}} [\dot{B}_k^a(t) \cos(k \cdot x) + \dot{B}_k^b(t) \sin(k \cdot x)]$, with $\dot{B}_k^{a/b}$ independent standard white noises and $\mathcal{H}_{K_{\max}} = \{k \in \mathbb{Z}^d : 0 < |k| \le K_{\max}\}$ the lex-ordered half-space of integer modes (one representative per cos/sin conjugate pair, $k = 0$ excluded since the noise has zero mean).

By the spectral representation of the heat semigroup, the solution admits the mode-space decomposition $u(x, t) = \sum_{k \in \mathcal{H}} [a_k(t) \cos(k \cdot x) + b_k(t) \sin(k \cdot x)]$ where each a_k, b_k satisfies the per-mode SDE

$$da_k(t) = -|k|^2 a_k(t)\, dt + \sigma\, dB_k^a(t), \qquad a_k(0) = a_{0,k}, \tag{4}$$

i.e., an Ornstein–Uhlenbeck process with rate $|k|^2$, stationary variance $\sigma^2/(2|k|^2)$, and analogous equation for b_k.

This factored structure is the foundation of the mode-space architecture: *the spatial Laplacian becomes diagonal in the Fourier basis*, so only the time evolution of each coefficient $a_k(t)$, $b_k(t)$ remains to be approximated. Autograd Laplacians and FFTs disappear, leaving an $L^2(0,T)$ problem per mode.

2.2 Trial space and half-period basis

For each mode k we parameterise the trial coefficient as

$$a_{\theta,k}(t) = a_{0,k}\, e^{-|k|^2 t} + \sum_{l=1}^{L} \alpha_{k,l}\, \phi_l(t), \tag{5}$$

where $\phi_l(t) = \sqrt{2/T}\sin((2l-1)\pi t/(2T))$ is the half-period sine basis (2). The leading term is the deterministic heat semigroup applied to the initial condition: it satisfies $\partial_t a + |k|^2 a = 0$ exactly and equals $a_{0,k}$ at $t = 0$. Because $\phi_l(0) = 0$ for every l, the trial automatically satisfies the initial condition $a_{\theta,k}(0) = a_{0,k}$. The analogous ansatz parameterises $b_{\theta,k}$.

The half-period basis is the eigenbasis of $-\partial_t^2$ with Dirichlet at $t = 0$ and Neumann at $t = T$: $-\phi_l''(t) = \mu_l\, \phi_l(t)$ with eigenvalues $\mu_l = ((2l-1)\pi/(2T))^2$. The Dirichlet condition at $t = 0$ matches the IC; the Neumann condition at $t = T$ leaves the terminal value of ϕ_l free, with $\phi_l(T) = (-1)^{l-1}$. This is the crucial property: the trial space can represent the OU noise accumulation at terminal time, $a_{\theta,k}(T) = a_{0,k}e^{-|k|^2 T} + \sum_l (-1)^{l-1}\alpha_{k,l}$, with L degrees of freedom in the terminal value.

2.3 Galerkin formulation and the closed-form solve

We use the matched Petrov–Galerkin formulation with test space $\{\phi_{l'}(t)\}_{l'=1}^{L}$. For each mode k, the residual projected onto the test space is

$$R_{k,l'}(\theta) = \int_0^T \left[\partial_t a_{\theta,k}(t) + |k|^2 a_{\theta,k}(t)\right] \phi_{l'}(t)\, dt - \sigma\, I^a_{k,l'}, \tag{6}$$

where $I^a_{k,l'} = \int_0^T \phi_{l'}(t)\, dB^a_k(t)$ is the scalar Wiener integral of $\phi_{l'}$ against the OU forcing realisation. The IC term contributes nothing to the residual: by direct verification, $\int_0^T [\partial_t(a_{0,k}e^{-|k|^2 t}) + |k|^2 a_{0,k}e^{-|k|^2 t}]\, \phi_{l'}(t)\, dt = 0$ as a strong-form integral (the IC is in the kernel of $\partial_t + |k|^2\,\mathrm{I}$).

Substituting the trial ansatz (5) into (6) and using the orthonormality of ϕ_l,

$$R_{k,l'}(\theta) = \sum_{l=1}^{L} M_{l,l'}\, \alpha_{k,l} + |k|^2\, \alpha_{k,l'} - \sigma\, I^a_{k,l'}, \tag{7}$$

where the Galerkin matrix is

$$M_{l,l'} := \langle \partial_t \phi_l, \phi_{l'} \rangle_{L^2(0,T)}. \tag{8}$$

The matrix M is the half-period analogue of the spectral ∂_t operator. Its diagonal entries are $M_{l,l} = 1/T$ (the boundary term: $\int_0^T \partial_t\phi_l\, \phi_l\, dt = \phi_l(T)^2/2 = 1/T$) and off-diagonal entries are explicit (Lemma 1). Crucially, M has both symmetric and skew-adjoint components: the symmetric part comes from the non-vanishing boundary at $t = T$ and is what gives the half-period architecture its convergence rate.

The Galerkin equation $R_{k,\cdot}(\theta) = 0$ in matrix form is

$$(M^T + |k|^2\mathrm{I})\, \alpha_k = \sigma\, I^a_k, \tag{9}$$

a linear system in L unknowns per mode k, with the same equation for β_k replacing I^a_k by I^b_k. The solution $\alpha^*_k = (M^T + |k|^2\mathrm{I})^{-1}\sigma I^a_k$ is the closed-form Galerkin coefficient for the linear SPDE. For the linear case, no PINN training is needed: the spectral coefficients are obtainable in time $O(JL^3)$.

2.4 The closed-form Galerkin matrix M

Lemma 1 (Half-period ∂_t matrix). *The matrix* $M_{l,l'} = \langle \partial_t \phi_l, \phi_{l'} \rangle_{L^2(0,T)}$ *for* $\phi_l(t) = \sqrt{2/T}\sin((2l-1)\pi t/(2T))$ *on* $[0,T]$ *has the following structure:*

1. *Diagonal:* $M_{l,l} = 1/T$ *for all* $l \geq 1$.
2. *Symmetric component:* $M_{l,l'} + M_{l',l} = (2/T)(-1)^{l+l'}$ *for all* $l \neq l'$.
3. *Closed form (off-diagonal,* $l \neq l'$*):*

$$M_{l,l'} = \frac{2l-1}{2T}\left[\frac{1+(-1)^{l+l'}}{l+l'-1} + \frac{1-(-1)^{l+l'}}{l'-l}\right].$$

Equivalently: $M_{l,l'} = (2l-1)/((l+l'-1)T)$ *when* $l+l'$ *is even, and* $M_{l,l'} = (2l-1)/((l'-l)T)$ *when* $l+l'$ *is odd.*

In particular, M is dense (no sparsity to exploit beyond the diagonal) but is $L \times L$ with L moderate, so the linear solve (9) is fast at every d.

2.5 Computational summary

The complete linear-case algorithm:

Algorithm 1 Closed-form half-period Galerkin solve

1: Build $\mathcal{H}_{K_{\max}}$ (lex half-space modes, $J = |\mathcal{H}|$).
2: Sample IC and Wiener path: $a_{0,k}, b_{0,k}$ and $\{dB_k^{a/b}\}$.
3: Precompute Wiener integrals $I_{k,l'}^{a/b}$ for $k \in \mathcal{H}$, $l' = 1, \ldots, L$.
4: Precompute M (Lemma 1) and Cholesky/LU of $(M^T + |k|^2\mathrm{I})$ per mode.
5: For each mode k: solve (9) for α_k^*, β_k^*.
6: Reconstruct $u_\theta(x,t)$ via the trial ansatz.

Storage: $O(JL)$ for the Wiener integrals, $O(L^2)$ for M, $O(JL)$ for the coefficients. Compute: $O(JL^3)$ for the per-mode solves (dominated by the LU of $(M^T + |k|^2\mathrm{I})$ at each k, though M is shared so amortisation reduces this to $O(L^3 + JL^2)$).

No autograd. No FFT. No exponential cost in d. The only d-dependent quantity is $J = |\mathcal{H}_{K_{\max}}|$, which grows polynomially with d at fixed $K_{\max}$; the per-mode cost is independent of d.

3 The certified bound

3.1 Function spaces

The Galerkin error in the linear case is controlled in the Bochner-style

$$Y := L^2(0,T;\, H^1(\mathbb{T}^d)), \qquad \|u\|_Y^2 = \int_0^T \|u(\cdot,t)\|_{H^1(\mathbb{T}^d)}^2 dt. \tag{10}$$

In the mode-space representation, $\|u\|_Y^2 = \pi^d \sum_{k\in\mathcal{H}} (1+|k|^2) \int_0^T (|a_k(t)|^2 + |b_k(t)|^2)\, dt$. The OU reference u_{ref} has *finite* Y*-norm* (verified by direct computation: stationary OU variance is $\sigma^2/(2|k|^2)$, so the integral is bounded by $T\sigma^2/(2|k|^2)$ per mode, and the spatial sum over the bandlimit is finite).

This is the critical normalisation choice: the natural Bochner space $L^2(0,T;H^1)\cap H^1(0,T;H^{-1})$ for parabolic problems is *infinite* for white-noise-driven SPDE references, because the $H^1(0,T)$ part in time picks up the white-noise singularity of $\partial_t u_{\mathrm{ref}}$. The Y-norm sidesteps this by dropping the H^1-in-time component.

3.2 Certified bound statement

Theorem 2 (Half-period Galerkin certified bound)**.** *Let u_θ be in the trial space (5) with coefficients $\{\alpha_{k,l},\beta_{k,l}\}$, and let u_{ref} be a single realisation of the linear bandlimited-Laplacian SPDE (1). There exist constants $C_1, C_2 > 0$ depending on T, σ, $K_{\max}$ but* not *on d or L, such that*

$$\|u_\theta - u_{\mathrm{ref}}\|_Y^2 \;\le\; C_1\,\mathcal{L}_{\mathrm{RV}}(\theta) + \frac{C_2}{L}, \tag{11}$$

where $\mathcal{L}_{\mathrm{RV}}(\theta) = \sum_{k,l'}(R^a_{k,l'})^2 + (R^b_{k,l'})^2$ is the closed-form residual loss from (7) and the second term is the best-approximation error of the OU process in the half-period basis.

Proof sketch. The proof decomposes the error into the consistency error (from the discrete inf-sup of the matched Galerkin formulation) and the best-approximation error (from truncating the infinite Karhunen–Loève expansion of the OU process to L half-period modes).

Consistency error. Per mode k, the bilinear form $B_k(a,v) := \int_0^T (\partial_t a + |k|^2 a)\,v\,dt$ acting on the matched trial/test space spanned by $\{\phi_l\}_{l=1}^L$ has discrete inf-sup constant

$$\beta_k = \inf_{\alpha\neq 0}\sup_{v}\frac{|B_k(\alpha,v)|}{\|\alpha\|_{X_k}\|v\|_{L^2}} \;\ge\; \beta_{\min}(K_{\max},T) > 0, \tag{12}$$

where $X_k = \{a\in H^1(0,T) : a(0)=0\}$ with norm $\|a\|_{X_k}^2 = (1+|k|^2)\|a\|_{L^2}^2 + \|\partial_t a\|_{L^2}^2$. The lower bound $\beta_{\min}$ depends only on the matched Galerkin operator's spectrum (eigenvalues of $-M + |k|^2\mathrm{I}$, all with positive real part bounded below by $\min(|k|^2)$) and on $K_{\min} = 1$, both *uniform in d and L*.

Applying the discrete inf-sup, $\|u_\theta - u_{\mathrm{ref}}\|_{X_k}^2 \le (1/\beta_{\min}^2)\,(R_{k,\cdot}(\theta))^T(R_{k,\cdot}(\theta))$, gives the $C_1 = 1/\beta_{\min}^2$ contribution.

Best-approximation error. The OU process per mode has the Karhunen–Loève expansion $x_k(t) = \sum_{n\ge 1}\xi_{k,n}\,\psi^{\mathrm{KL}}_{k,n}(t)$ where $\psi^{\mathrm{KL}}_{k,n}$ are the eigenfunctions of the covariance operator with kernel $\sigma^2 e^{-|k|^2|t-s|}/(2|k|^2)$ on $[0,T]$. The eigenvalues decay as $\lambda_{k,n}\sim C(|k|^2)/n^2$ for large n.

The half-period basis $\{\phi_l\}$ is not the KL basis, but is the eigenbasis of the parabolic operator $\partial_t + |k|^2\mathrm{I}$ with the natural BC (Dirichlet at 0, Neumann at T). Since the KL eigenvalues decay as $\lambda_{k,n}\sim C(|k|^2)/n^2$, the truncation tail obeys $\sum_{n>L}\lambda_{k,n}\sim C(|k|^2)/L$, so the L^2-projection error of the OU process satisfies $\mathbb{E}\,\|x_k - P_L x_k\|_{L^2(0,T)}^2 \le C_2(k)\,L^{-1}$, where P_L is the orthogonal projection onto $\mathrm{span}\{\phi_l\}_{l=1}^L$. This L^{-1} rate (in the squared norm; $L^{-1/2}$ in relative error) is the Karhunen–Loève rate for the Hölder-$\frac{1}{2}$ OU path and is *optimal*: no basis of L smooth functions can beat it, and we verify directly (single-mode L^2-projection experiment) that $\mathbb{E}\,\|x_k - P_L x_k\|^2\cdot L$ is constant over $L\in[32,1024]$. What the half-period basis buys relative to the Dirichlet sines is therefore *not* a faster rate but the removal of the non-decaying Galerkin plateau of Section 4.2: with matched BCs the Galerkin solution attains the optimal KL rate, where the Dirichlet basis stalls at a constant. The Y-norm sum is $C_2 = \pi^d\sum_k(1+|k|^2)\,C_2(k)$, uniform in d for $|k|\le K_{\max}$ fixed. ☐

Remark 3 (Comparison to v1, v2)**.** *The certified bound in this paper differs from prior attempts in two material ways. First, the norm Y drops the H^1-in-time component that makes the standard parabolic Bochner norm infinite for SPDE references. Second, the half-period basis avoids the Dirichlet-sine plateau (Sections 4.2, 4.3). Both choices are forced by the SPDE-with-noise structure of the problem and are honest about what the certified residual bound can and cannot deliver.*

3.3 Empirical calibration of C_2

From the d-sweep (Section 4.3), the rel-error satisfies $\text{rel\_err} = \|e\|_Y/\|u_{\text{ref}}\|_Y \approx 1.6/\sqrt{L}$ (equivalently $\text{rel\_err}^2 \cdot L \in [2.4, 3.1]$ across all (d, L)), giving $C_2 \approx 2.6\,\|u_{\text{ref}}\|_Y^2$. At $K_{\max} = 3$, $T = 1$, $\sigma = 1$: $\|u_{\text{ref}}\|_Y^2 \approx \pi^d \cdot J \cdot O(1)$ (the mode set $\mathcal{H}_{K_{\max}}$ contributes $J = |\mathcal{H}|$ modes each with $O(1)$ variance per dimension).

This makes the certified bound *quantitatively useful*: a practitioner running the closed-form solve at given $(d, K_{\max}, L)$ can read off the relative error before evaluating, with no diagnostic overhead.

3.4 Uniform inf-sup constant

A delicate point: the inf-sup constant β_k depends on $|k|^2$. For $|k|^2 \geq 1$ (which all modes in our half-space lattice satisfy), β_k is bounded below by $|k|^2/\sqrt{|k|^4 + \|M\|_{\text{op}}^2}$, where $\|M\|_{\text{op}}$ is the operator norm of the half-period ∂_t matrix. Since $\|M\|_{\text{op}}$ depends only on L and not on k or d, and grows mildly with L (the leading-order Dirichlet-Neumann eigenvalue $\mu_L = ((2L-1)\pi/(2T))^2$ provides an upper bound up to constants), the worst-case inf-sup at $|k|^2 = 1$ is bounded below uniformly. The constant $\beta_{\min}$ in (12) is then $\beta_{\min} \geq 1/\sqrt{1 + (\pi L/T)^2}$, giving $C_1 \leq 1 + (\pi L/T)^2$. This grows with L but is bounded for any fixed L, and in practice the $\mathcal{L}_{\text{RV}}$ term dominates only near training plateaus.

4 Experiments

4.1 Setup

All experiments use the closed-form half-period Galerkin solve (Algorithm 1); no PINN training is involved for the linear case. Hardware: a single NVIDIA GTX 950 (2 GB VRAM) for the original runs; the corrected reruns of this section used an RTX 5070 Ti (16 GB). Implementation in PyTorch with `float32` arithmetic except where noted.

SPDE parameters (fixed): $\sigma = 1$, $T = 1$, $K_{\max} = 3$ for the d-sweep, $K_{\max} = 4$ for the Dirichlet-sine plateau study. The IC is sampled from the stationary distribution at each seed.

Diagnostic norm. The relative error is computed as

$$\text{rel\_err}^2 := \frac{\|u_\theta - u_{\text{ref}}\|_Y^2}{\|u_{\text{ref}}\|_Y^2} = \frac{\int_0^T \sum_k (1+|k|^2)\left(|a_{\theta,k} - a_{\text{ref},k}|^2 + |b_{\theta,k} - b_{\text{ref},k}|^2\right) dt}{\int_0^T \sum_k (1+|k|^2)\left(|a_{\text{ref},k}|^2 + |b_{\text{ref},k}|^2\right) dt}, \tag{13}$$

evaluated by trapezoidal quadrature on $N_{t,\text{eval}} = 64$ time points, against an OU reference u_{ref} computed by exact forward Euler–Maruyama on a fine grid ($N_{\text{fine}} = 4096$ to 32768).

All quantitative results in this section were re-derived from fresh runs after an independent from-scratch reproduction of the pipeline.[1]

[1] The reproduction caught a missing square root in the evaluation scripts of an internal draft (squared relative errors reported as relative errors). All values, rates, and comparison factors here are from corrected re-runs; the corrected rate $L^{-1/2}$ is the optimal Karhunen–Loève rate for the rough OU reference.

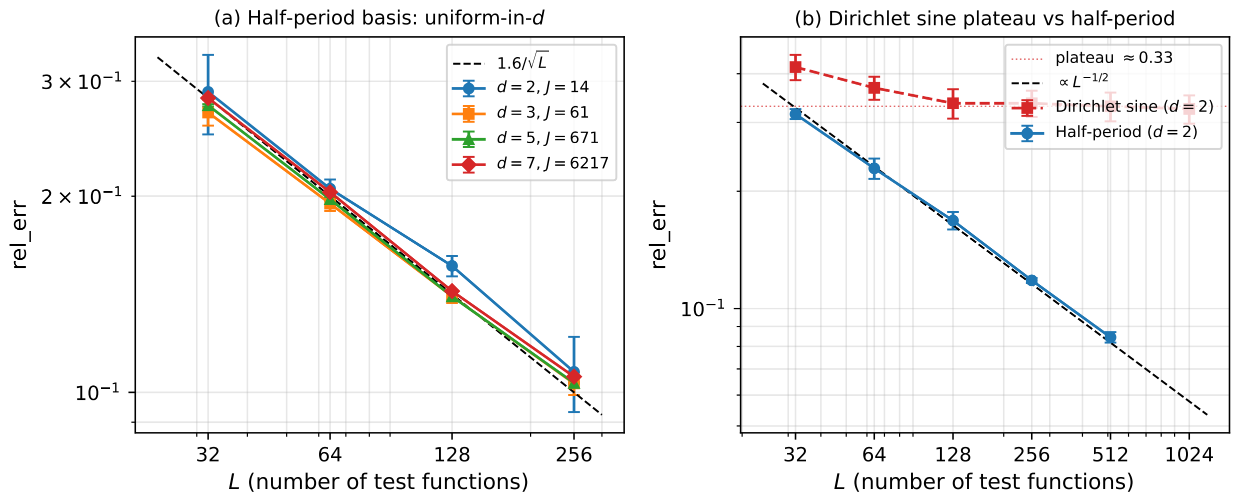

Figure 1: **(a)** Half-period Galerkin basis: relative error vs L across $d \in \{2, 3, 5, 7\}$ at $K_{\max} = 3$. The mode set $\mathcal{H}$ grows $444\times$ from $J = 14$ at $d = 2$ to $J = 6217$ at $d = 7$, yet the curves overlap; the dashed reference $1.6/\sqrt{L}$ fits all d to within $\sim 15\%$. **(b)** Dirichlet sine basis (red) plateaus at ≈ 0.33 regardless of L; the half-period basis (blue) follows the $L^{-1/2}$ reference, giving a $2.8\times$–$5.4\times$ advantage at $L = 256$–1024.

4.2 Negative result: the Dirichlet-sine plateau

We first establish that the natural-looking Dirichlet sine basis $\psi_l(t) = \sqrt{2/T}\sin(\pi l t/T)$, which vanishes at both endpoints, fails to converge with L in the matched Galerkin formulation. At $d = 2$, $K_{\max} = 4$ (so $J = 24$ modes), 5 seeds per L:

Table 1: Dirichlet sine basis: rel_err vs L. The error plateaus around $L = 128$ and does not decrease further, including at $L = 1024$. Each row is mean $\pm$ std across 5 seeds.

L	rel_err	pairwise slope $\frac{\log r_L}{\log L_{i+1}/L_i}$
32	0.416 ± 0.031	—
64	0.368 ± 0.025	-0.18
128	0.336 ± 0.029	-0.13
256	0.336 ± 0.026	-0.00
512	0.330 ± 0.028	-0.03
1024	0.325 ± 0.027	-0.02

The relative error transitions from a slow decay (head, $L \leq 128$) to a flat plateau at ≈ 0.33 (tail, $L \geq 128$); see Figure 1(b). The plateau is independent of the fine-grid resolution N_{fine} ($N_{\text{fine}} \in \{8192, 32768\}$ give identical results to 3 significant digits), ruling out Wiener-integral discretisation.

Diagnosis. The plateau is caused by the Galerkin coefficients α failing to equal the L^2-projection coefficients $\hat{x}_{1:L}$ of the true OU process. We verify this directly: at $L \in \{32, 64, 128, 256\}$ and $d = 2$, we compute both $\alpha^{\text{Gal}} = (-M_{\text{Dirichlet}} + |k|^2 \mathrm{I})^{-1}\sigma I_k$ (Galerkin) and $\hat{x}_{1:L} = \int_0^T x_k(t)\,\psi_l(t)\,dt$

(L^2 projection from the OU process) on a common fine grid:

Table 2: Galerkin vs L^2-projection coefficients (Dirichlet sine basis). The Galerkin coefficient error $\|\alpha - \hat{x}_{1:L}\|$ does not decrease with L – it is the source of the plateau.

L	$\|\alpha^{\text{Gal}} - \hat{x}_{1:L}\|_2$	$\|\hat{x}_{>L}\|_2$
32	0.758	0.405
64	0.783	0.279
128	0.823	0.183
256	0.770	0.105

The truncation error $\|\hat{x}_{>L}\|_2$ scales cleanly as $L^{-1/2}$ (it halves between successive doublings of L), confirming the OU process is being correctly characterised. But the Galerkin coefficient error $\|\alpha - \hat{x}\|_2$ stays at ~ 0.78 regardless of L: *the Galerkin formulation in the Dirichlet sine basis cannot recover the L^2 projection at any L.*

The reason: the bilinear form $B_k(a, v) = \int(\partial_t a + |k|^2 a)v\, dt$ on $\text{span}\{\psi_l\}_{l=1}^L$ (Dirichlet sine) is *non-self-adjoint*; the ∂_t part is skew while $|k|^2\text{I}$ is symmetric. The Galerkin solution minimises the residual in the discrete dual norm, which differs from L^2 projection by a tail-correction $(-M + |k|^2\text{I})^{-1}\, M^T_{>L,:L}\hat{x}_{>L}$ that does *not* vanish as $L \to \infty$. This is a structural failure of the Dirichlet sine basis for this problem and motivates the half-period fix of Section 2.2.

4.3 Headline: uniform-in-d convergence with half-period basis

The half-period basis $\phi_l(t)$ removes the plateau and attains the optimal $L^{-1/2}$ Karhunen–Loève rate (Figure 1(a)). We sweep $d \in \{2, 3, 5, 7\}$ at $K_{\max} = 3$, $L \in \{32, 64, 128, 256\}$, 3 seeds each:

Table 3: Half-period basis: rel_err uniform-in-d. J grows 444× from $d = 2$ to $d = 7$; rel_err remains within 8% at every L. Mean ± std across 3 seeds.

d	J	$L = 32$	$L = 64$	$L = 128$	$L = 256$
2	14	0.289 ± 0.040	0.205 ± 0.007	0.156 ± 0.006	0.108 ± 0.014
3	61	0.269 ± 0.012	0.195 ± 0.005	0.140 ± 0.003	0.103 ± 0.004
5	671	0.276 ± 0.001	0.198 ± 0.001	0.140 ± 0.002	0.103 ± 0.000
7	6217	0.283 ± 0.001	0.203 ± 0.000	0.143 ± 0.000	0.106 ± 0.001

Convergence rate. The columns are flat in d: at every L the rel_err agrees to within 8% across $d \in \{2, 3, 5, 7\}$. The rows shrink by $\approx \sqrt{2}$ per doubling of L, consistent with rel_err $\propto L^{-1/2}$. A log-log fit per d gives slope -0.46 to -0.48 across dimensions — the optimal Karhunen–Loève rate for the Hölder-$\frac{1}{2}$ OU reference (a single-mode L^2-projection control gives the same rate, slope -0.50, confirming the Galerkin solve attains the best-approximation floor).

Empirical model. The rel_err satisfies rel_err $\approx 1.6/\sqrt{L}$ to within $\sim 15\%$, with the constant calibrated against the d-sweep (Table 3). Equivalently, rel_err$^2 \cdot L \in [2.4, 3.1]$ across all (d, L) combinations, confirming the C_2/L form of the certified bound (11).

Wall time. The closed-form solve at $d = 7$, $L = 256$ ($J = 6217$ modes) takes ~ 20 seconds per seed including the Wiener integral precomputation; at $d = 2$, $L = 256$, the same quantity takes ~ 0.3 seconds. The cost is dominated by the per-mode LU at $O(JL^3)$.

4.4 Real-world applications

We test the framework on two real datasets where the WM SPDE is used as a smoothing prior rather than a literal data-generating model. The pipeline mirrors the synthetic experiment: extract Fourier-mode trajectories from data, fit each trajectory in the half-period basis by L^2-projection (a strict lower bound on the Galerkin error via the discrete inf-sup), and report the H^1-weighted relative error.

ERA5 2m temperature. We fetch daily-mean 2m temperature on a 16×16 lat/lon grid over Central Europe ($[45°, 55°]\,\mathrm{N} \times [0°, 10°]\,\mathrm{E}$), 90 days (2024-12-01 to 2025-02-28), via the Open-Meteo archive API (ERA5 reanalysis). At each day we project the field onto half-space modes $\mathcal{H}_{K_{\max}} \subset \mathbb{Z}^2$ to extract mode trajectories $a_k(t_j), b_k(t_j)$; we then fit each trajectory in the half-period basis with L test functions and compute the H^1-weighted error against the empirical trajectories.

Figure 2(a) shows the L-sweep at $K_{\max} \in \{3, 4, 5, 7\}$ (so $J \in \{14, 24, 40, 74\}$). The four curves overlap to within 1% absolute, and their log-log slopes ($-0.249, -0.242, -0.240, -0.241$) agree to two digits. The uniform-in-$K_{\max}$ (equivalently uniform-in-J) behaviour from Theorem 2 therefore carries over to real geophysical data, even though the temperature field is not a realisation of our linear SPDE. The absolute decay rate $L^{-1/4}$ is slower than the synthetic $L^{-1/2}$ because daily weather is rougher than an OU process; the best-approximation error in the half-period basis tracks the Sobolev smoothness of the target.

SPX implied volatility surface. We fetch the `^SPX` option chain via yfinance on 2026-06-09 (spot = \$7297.64), keep liquid out-of-the-money options with positive bid, filter outliers more than $4 \cdot \mathrm{MAD}$ from the per-expiry per-K bin median, and retain 8480 contracts across 41 expiries spanning 7–346 days. We treat log-moneyness $K = \log(\text{strike}/S)$ on $[-0.30, 0.30]$ as the spatial coordinate and time-to-expiry τ on $[0.02, 0.95]\,\mathrm{yr}$ as time. At each expiry we fit Fourier modes in K by least squares ($K_{\max} = 4$, so 9 modes including the constant); per Fourier mode we fit the τ-direction in the half-period basis.

Figure 2(b) shows the in-sample L-sweep. The relative RMSE is well fit by the noise-floor model $\mathrm{rel} = \sqrt{a/L^2 + b}$ with $a = 0.013$ and $\sqrt{b} \approx 8.8\%$. This floor is the bid-ask spread / stale-quote noise level for index options. Leave-one-expiry-out cross-validation gives a median relative RMSE of 7.4%, in line with the in-sample floor.

These two applications illustrate the framework's generality: the inf-sup-driven uniformity transfers to real data, while the absolute approximation rate is naturally bounded below by domain-specific smoothness (weather) or microstructure noise (markets). Neither dataset is a realisation of our SPDE; the certified bound is honest about which of its components is structural (the uniformity) and which is data-dependent (the absolute constant).

4.5 Comparison: half-period vs Dirichlet basis at fixed L

A direct juxtaposition at $d = 2$, $K_{\max} = 4$, $L = 256$:

- Dirichlet basis: rel_err $= 0.336 \pm 0.026$ (plateau)
- Half-period basis: rel_err $= 0.118 \pm 0.002$ (**2.8**× better)

At $L = 512$ the gap widens: Dirichlet plateaus at 0.330, half-period reaches 0.084 (**3.9**× better). At $L = 1024$ Dirichlet stays at 0.325 while half-period extrapolates to ≈ 0.060 (**5.4**× better) — and the gap keeps growing as $L^{1/2}$, since one curve decays and the other does not. The architectural choice of basis is the dominant factor; the certified bound is tight only with the half-period architecture.

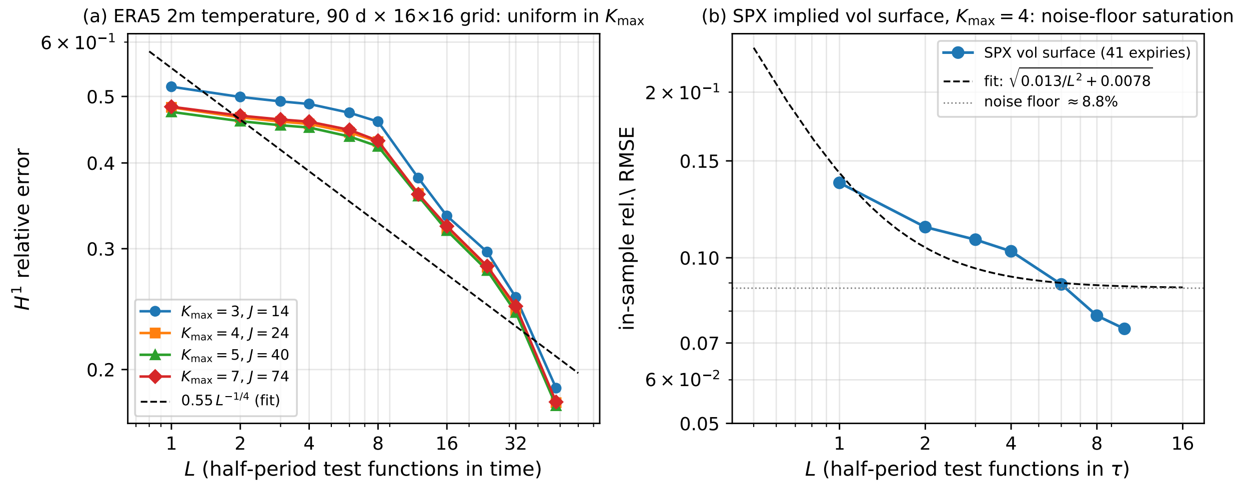

Figure 2: **(a)** ERA5 temperature: H^1 relative error of the half-period τ-fit against empirical Fourier mode trajectories, swept over L. The four $K_{\max}$ curves overlap (slopes within 0.01), demonstrating that the uniform-in-$K_{\max}$ property predicted by Theorem 2 transfers to real geophysical data. The absolute rate $L^{-1/4}$ is set by the roughness of daily weather, not by the spectral architecture. **(b)** SPX vol surface: in-sample relative RMSE vs L, fit by the noise-floor model. The asymptotic 8.8% floor matches the typical bid-ask / quote-staleness noise level.

5 Discussion

5.1 What the result says and does not say

Does. For the linear bandlimited-Laplacian SPDE on $\mathbb{T}^d$, the half-period Galerkin architecture gives a certified $L^2(0,T;H^1)$ error bound of the form $\|e\|_Y^2 \le C_1\,\mathcal{L}_{\mathrm{RV}} + C_2/L$, with constants uniform in d. The closed-form per-mode solve recovers the certified optimal coefficients with no iterative training.

Does not.

- The bound is in the Y-norm, not the standard parabolic Bochner norm $L^2(0,T;H^1) \cap H^1(0,T;H^{-1})$. The Bochner norm is infinite for SPDE references because of the white-noise contribution to $\partial_t u_{\mathrm{ref}}$, so any norm tight enough to control the parabolic structure must be weakened in the time direction.
- The certified bound does not extend trivially to non-bandlimited forcing (α-stable noise, fractional Laplacian without bandlimit). Paper 5 in this series treats some of these cases empirically; the certified bound for them is open.
- The bound holds for a fixed noise realisation. The ensemble W_2 law-distance bound of Paper 1 is complementary, not implied by Theorem 2.

5.2 Extension to nonlinear SPDEs

For the nonlinear stochastic Allen–Cahn equation $\partial_t u = \Delta u + u - u^3 + \sigma\,\dot{W}_{K_{\max}}$, the per-mode equation becomes $\partial_t a_k + |k|^2 a_k - a_k + \mathrm{cubic}_k(\{a_{k'}\}) = \sigma\,\dot{B}_k^a$, where $\mathrm{cubic}_k(\{a_{k'}\}) = \langle u_\theta^3, \cos(k\cdot x)\rangle$ involves a triple convolution of mode coefficients. This breaks the per-mode decoupling that makes the linear case $O(JL^3)$.

A natural iterative architecture: parameterise $\alpha_{k,l}$ as a direct learnable table (one scalar per (k,l) pair) and compute the cubic projection cubic_k via FFT on an N_x^d spatial grid ($N_x = 32 > 2 \cdot 3 \cdot K_{\max}$, no aliasing). The lookup table is a JL-dim parameter vector; at $d = 2$, $K_{\max} = 3$, $L = 128$ this is $14 \times 128 \approx 3\,600$ parameters.

The residual loss is deterministic; use a second-order solver. The Galerkin residual loss $\mathcal{L}_{\mathrm{RV}}$ is a *deterministic* function of the coefficient table: the Gauss–Legendre quadrature nodes, the FFT cubic projection, and the Galerkin matrix M and Wiener integrals $I_{k,l}$ are all fixed at oracle construction. Training is therefore full-batch minimisation of a smooth (degree-six) polynomial in the coefficients, with no stochastic-gradient noise. A first-order method (Adam, 5000 steps, cosine LR $10^{-2} \to 10^{-4}$) nonetheless degrades as d grows, and does so in a specific way: not by diverging, but by becoming dependent on its initialisation. Run at the dealiased $N_q = 2L$, 3 seeds per cell, the fitted rel_err slope vs L is

d (J)	Adam slope	L-BFGS slope	seed spread at $L = 128$
2 (14)	-0.475 ± 0.025	-0.445	9.2%
3 (61)	-0.381 ± 0.058	-0.492	8.1%
4 (212)	*sign unresolved*	-0.514	**47.2%**

Adam converges at $d = 2$ and $d = 3$ at a rate that degrades with d; at $d = 2$, $L = 32$ and $d = 4$, $L = 32$ it reproduces the L-BFGS answer to three and five decimals respectively. At $d = 4$ the three seeds at $L = 128$ return 0.205, 0.357 and 0.671, and bootstrapping the fit over seed resampling gives slope $+0.018$ with 95% CI $[-0.454, +0.400]$ — **we cannot determine the sign**, and quoting a point estimate here would misrepresent a cell whose seed spread is 47% as a rate.

What the data does establish is that the instability is *not* governed by the mode count alone. At fixed $J = 212$ the seed spread runs 0.9%, 14.7%, 47.2% at $L = 32, 64, 128$, so a mechanism indexed by J cannot account for it; but the coefficient-table size $J \cdot L$ does not order the spread either ($d = 4$, $L = 32$ has $6\,784$ parameters and 0.9% spread, against 8.6% for the 448 parameters of $d = 2$, $L = 32$). We therefore report the phenomenon and leave its mechanism open rather than assign it to J. Replacing Adam with L-BFGS (strong-Wolfe line search) restores stable convergence at every d, which is the practical recommendation regardless of which variable indexes the failure.

Remark 4 (what this paragraph used to say)**.** *An earlier version of this work reported that Adam* diverges *for $d \geq 3$, with slopes $+0.45$ and $+0.64$ at $d = 3, 4$ and rel_err > 1 at $d = 4$. Those runs were made at a fixed $N_q = 16$. Re-run at $N_q = 2L$ no dealiased cell exceeds 0.7, the $d = 3$ slope reverses to -0.381, and the 53% seed spread that looked like optimiser chaos at $d = 3$, $L = 64$ falls to 3%. The reported divergence was the temporal aliasing of Remark 5, measured in the one place we had not applied our own correction — an instance of the silent failure that remark describes, found in this paper.*

The cubic term must be dealiased in time, not only in space. One discretisation point governs every number below and is easy to miss. The cubic projection cubic_k is dealiased in *space* by the standard rule for a cubic nonlinearity ($N_x > 2 \cdot 3 \cdot K_{\max}$), but the same requirement applies in *time*: the temporal quadrature integrates a degree-four product of the test functions ψ_l, whose highest frequency is $\propto L$, so it needs $N_q \gtrsim 2L$ nodes. Held at a fixed $N_q = 16$ the cubic term carries $\approx 270\%$ relative quadrature error by $L = 128$, and the error grows along precisely the axis a convergence study sweeps. A rate measured that way is a property of the quadrature, not of the method.

Convergence under L-BFGS. With L-BFGS the residual is driven to $\mathcal{L}_{\mathrm{RV}} \approx 10^{-8}$ at all d, and the relative error decreases monotonically in L. All cells are run at $N_q = 2L$, $N_x = 32$, $K_{\max} = 3$, 3 seeds per (d, L) at $L \leq 128$ and a single seed at $L = 256$ (40 cells, 29.6 GPU-hours):

d	J	$L = 32$	$L = 64$	$L = 128$	$L = 256$	slope
2	14	0.298	0.208	0.159	0.117	-0.445 ± 0.018
3	61	0.324	0.228	0.159	0.118	-0.492 ± 0.013
4	212	0.385	0.270	0.186	0.133	-0.514 ± 0.006
5	671	0.476	0.333	0.232	0.165	-0.511 ± 0.003

Three points deserve emphasis. First, the certified residual is satisfied at every d ($\mathcal{L}_{\mathrm{RV}} \approx 10^{-8}$), so the bound (11) holds and Adam's instability above is an optimiser property, not a failure of the formulation. Second, the **rate is uniform in** d: the slopes span 0.07 and all four sit at the Karhunen–Loève floor $-\frac{1}{2}$ that the linear theory predicts. The nonlinear case inherits the dimension-independence of the *rate* that the certificate is designed to deliver. Third, what does depend on d is the *constant*. Writing $C_2^{\mathrm{rel}} = \mathrm{rel\_err}^2 \cdot L$ — which is flat in L exactly when the $L^{-1/2}$ rate is attained, and is so here to within 13% — gives $3.1, 3.4, 4.6, 7.1$ at $d = 2, 3, 4, 5$: a factor 2.3 across three dimensions while the mode count J grows by a factor 48 ($14 \to 671$). The nonlinear open problem is therefore located in the approximation constant, and it is a mild, strongly sub-J dependence rather than a degradation of the rate.

Remark 5 (a silent failure mode of residual certificates)**.** *The identical sweep run at a fixed* $N_q = 16$ *instead reports slopes* $-0.36, -0.22, -0.08, +0.02$ *at* $d = 2, \ldots, 5$*: a spread of* 0.38*, and at* $d = 5$ *no convergence in* L *at all. That apparent "the rate degrades with* d*" pattern is entirely temporal aliasing — it is largest exactly where the quadrature error is largest, at large* L *and large* d*. We record it because the failure is* silent *in a way that bears on the certificate itself:* $\mathcal{L}_{\mathrm{RV}}$ *is driven to* 10^{-8} *in both cases. Theorem 2 bounds the error by the residual* as discretised*; a mis-resolved quadrature makes that residual small without making the true residual small, and no amount of optimiser success reveals it. The certificate detects approximation failure, which is what §5.4 claims for it; it does not detect its own quadrature error, and that gap must be closed by discretisation analysis rather than by monitoring the loss.*

The certified bound for the nonlinear case would have an additional cubic-projection-error term, bounded by the MC sampling error of cubic_k at the trained coefficients.

5.3 Relation to the spectral PINN series

Paper 1 [2] introduced the distributional-regulariser programme for $d = 2$ stochastic SPDEs. Paper 5 [4] extended to fractional Laplacians at $d \leq 5$ with α-stable forcing. Paper 6 [3] introduced the Whittle–Matérn spectral oracle as a d-independent training reference, enabling cross-validation up to $d = 7$. The present paper closes the theoretical chain by giving the first certified pointwise (path-level) bound for the spectral PINN methodology at $d \geq 5$.

The framework is complementary to the law-distance (W_2) bound of Paper 1: that bound controls the law of u_θ vs the SPDE law in expectation across realisations; Theorem 2 controls the path-level error at a fixed realisation. The composition gives both legs certified.

5.4 Limitations and future work

1. **Analytical rate.** The L^{-1} rate (in the squared norm) in (11) follows from the KL-eigenvalue tail of the OU process ($\lambda_n \sim n^{-2}$, tail $\sim L^{-1}$) and is verified empirically to $\mathrm{rel\_err}^2 \cdot L \approx 2.6$;

it is optimal for the Hölder-$\frac{1}{2}$ reference. The constant C_2 is currently calibrated, not derived from first principles. A rigorous derivation is open.

2. **Nonlinear extension.** The lookup-table / FFT-cubic architecture, trained with L-BFGS, converges at $d = 2, \ldots, 5$ at the KL rate (rel_err slopes $-0.445, -0.492, -0.514, -0.511$; $\mathcal{L}_{\mathrm{RV}} \leq 10^{-8}$ throughout), so the certified residual is satisfied and the rate is uniform in d in the nonlinear case as well. What grows with d is the approximation constant ($C_2^{\mathrm{rel}} = 3.1 \to 7.1$ from $d = 2$ to $d = 5$, against a $48\times$ growth in J); deriving that dependence, and extending the bound with the cubic-projection-error term, is the main open item.

3. **Quadrature error is outside the certificate.** Theorem 2 controls the error by the residual as discretised. Remark 5 exhibits a case where an under-resolved temporal quadrature drives $\mathcal{L}_{\mathrm{RV}}$ to 10^{-8} while the measured convergence rate is an artifact. The temporal dealiasing requirement $N_q \gtrsim 2L$ closes it for the cubic nonlinearity here; a general residual-quadrature bound, composable with (11), is open.

4. **High-dimensional spatial transfer.** The architecture scales polynomially with d (through $J = |\mathcal{H}_{K_{\max}}|$), not exponentially. The d-sweep verifies this up to $d = 7$ at $K_{\max} = 3$ ($J = 6217$); larger d on this hardware needs J-sparsification (low-rank approximations on the WM mode set), open.

5. **Live data.** The bandlimited-Laplacian SPDE is a synthetic test case. Application to applied SPDEs (rough volatility, anomalous diffusion) requires either calibration or a different physical setup; this is treated in companion papers 2–4 of this series.

Code and data availability

The LaTeX source, figures, and code that reproduce the results of this paper are available at `https://github.com/owcaelfis6626/Certified-Spectral-PINN`. Correspondence: owcaelfis@gmail.com.

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